Smart Home Management System (Smart Home+)

SOFTWARE REQUIREMENTS SPECIFICATION (SRS) & DESIGN DOCUMENTATION

VERSION 1.0

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1. INTRODUCTION

1.1 Background to the Problem

In most regular homes, devices like lights, fans, air conditioners, security locks, and other electrical appliances are controlled manually. These devices usually work independently and rarely communicate with each other, which leads to common problems:

- **Energy waste:** People often forget to turn off lights or appliances when leaving home, which increases electricity bills.
- **Safety risks:** Appliances like heaters, or gas stoves left on accidentally can cause fire or gas leaks.
- **Limited remote control:** Users cannot check the status of devices when they are away from home.
- **Weak security monitoring:** Traditional security relies mostly on physical locks or separate alarms that do not send remote alerts.
- **Poor energy awareness:** Traditional homes do not provide information about electricity usage, making it difficult for users to track or reduce consumption.

Modern urban life has made smart home management even more necessary. Families often spend long hours outside for work, school, or travel, leaving homes unattended for long periods. Elderly family members or children staying at home also need monitoring and support. In such cases, users need a simple way to check the home, control devices remotely, and get alerts for unusual events. However, most homes still lack a centralized system that manages all devices, users, and security features together.

Target Audience: The proposed system is mainly designed for people who want an easier, safer, and more organized way to manage their homes using modern technology. Different groups of users can benefit from this system because each of them has different needs and responsibilities. My targeted audience:

Homeowners: The primary target audience of the system is homeowners who are responsible for managing household safety, electricity consumption, and device control. In traditional homes, homeowners often face difficulties in monitoring appliances when they are outside or busy with work. Many homeowners remain concerned about whether devices are turned off, doors are locked, or any unusual event has occurred at home.

Due to increasing work pressure and travel frequency, homeowners require a convenient way to stay aware of home conditions even when they are away. Therefore, they form the main user group because they need better visibility, control, and awareness of their home environment.

Family Members: Family members living in the same household are another important target audience. Family members regularly interact with home appliances such as lights, fans, air conditioners, and other electrical devices. In traditional homes, family members often face difficulties in safely operating devices without accidentally changing important settings or automation schedules.

Children, elderly family members, or less tech-savvy users may find it challenging to manage multiple devices simultaneously, which can sometimes lead to misuse or errors. Therefore, family members are provided with restricted access, allowing them to perform basic tasks such as turning devices on or off, while critical controls and system configurations remain under the homeowner’s authority.

This approach ensures that family members can use the system conveniently in their daily routines without compromising home safety or system integrity. At the same time, it provides peace of mind to homeowners, knowing that important settings and automation rules cannot be accidentally altered.

Guests: Guests are another target audience, have limited access to selected devices. In traditional homes, when guests come to a house while the homeowner is not present, they often have to wait in front of the home or ask the homeowner to operate devices like lights, fans, or air conditioners, which can be inconvenient and time-consuming. Using this system, the homeowner can give limited access to guests, allowing them to use certain appliances themselves for comfort during their visit, while still not having full control over the home.

For example, guests can turn allowed devices on or off, but cannot change automation schedules, access door locks or security cameras, or modify system settings. This keeps the homeowner’s control, security, and privacy intact while making it easier for guests to use the home comfortably.

Technicians: Technicians and maintenance personnel are also important targated audience because modern homes rely on electronic and network-connected devices that need proper setup, configuration, and occasional troubleshooting. In traditional homes, whenever a device breaks down or requires maintenance, homeowners often have to contact a technician, wait for an appointment, and sometimes even be physically present during the entire process. This can be time-consuming, inconvenient, and stressful for homeowners.

With the system, technicians can access the necessary tools and interfaces to install, configure, and maintain devices efficiently. They can ensure that appliances, security systems, and network connections are properly set up and functioning as intended. This directly benefits homeowners by reducing downtime, preventing errors, and ensuring that all devices work reliably without the homeowner needing to manage every detail themselves.

1.2 Selection of Process Model

Process Model: Incremental Process Model

The Incremental Process Model is a software development approach where the system is developed and delivered in multiple smaller parts called increments. Instead of building the entire system at once, the software is divided into manageable modules, and each module is designed, developed, tested, and delivered separately.

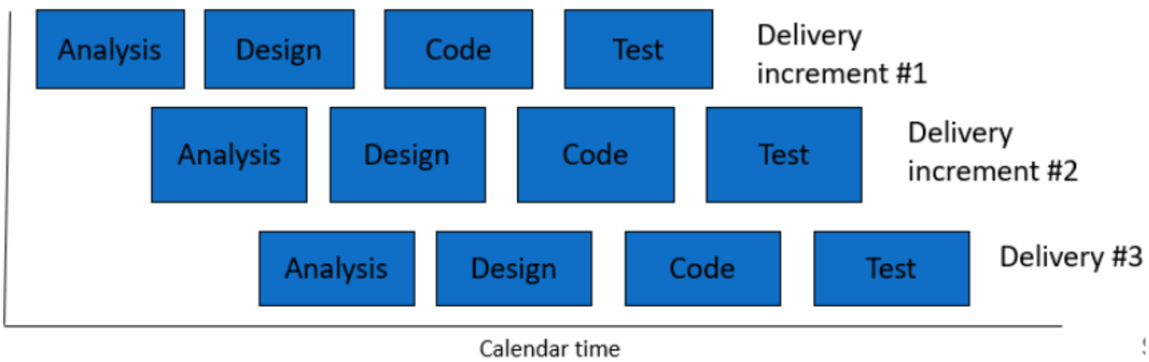
In this model, the first increment usually contains the most essential features of the system. After that, additional functionalities are added gradually in the next increments until the full system is completed.

For the Smart Home Management System, this approach allows the development team to first deliver the basic features such as user registration, login, and dashboard management. Then, more advanced modules such as device control, security monitoring, guest access, technician management, and energy tracking can be added step by step.

Each increment follows the same software development stages:

- Requirement Analysis
- System Design
- Implementation (Coding)
- Testing
- Deployment

This ensures that every module is fully functional before moving to the next stage.



Increment 1: Basic User Management

- Registration
- Login
- Homeowner Dashboard
- Family Member Dashboard.

Increment 2: Device Control

- Add smart devices
- Control lights, fans, AC
- Monitor device status

Increment 3: Security Module

- Smart door locks
- Security alerts
- Camera monitoring
- Emergency notifications

Increment 4: Maintenance

- Technician dashboard
- Contact homeowners.
- Device health check.
- Working status set.

Why I Chose Incremental Process Model

I selected the Incremental Process Model for my Smart Home Management System because the project is a mobile application that contains multiple interconnected modules such as user management, device control, security monitoring, voice assistant support, guest access, technician management, and energy tracking.

Developing all these functionalities at once would make the system highly complex, difficult to manage, and harder to test. Therefore, the Incremental Model is more suitable because it allows the project to be divided into smaller functional modules, where each module is developed, tested, and integrated step by step.

For example, the system can first implement user registration, login, and dashboard management. In the next increment, device control features such as lights, fans, and air conditioners can be added. After that, security functions including smart locks, alerts, and monitoring can be integrated.

Another reason for choosing this model is easier testing and debugging. Since each increment is tested separately, errors can be identified and resolved more efficiently without affecting the entire application.

Additionally, because this is a mobile-based smart home system, system stability and user experience are very important. Incremental development ensures that every module is functional and reliable before adding the next one, reducing development risk and improving software quality.

Therefore, the Incremental Process Model is the most appropriate choice for this project because it supports modular development, easier testing, better project management, reduced complexity, and reliable integration of multiple smart home features within a mobile application.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS

2.1 user Story Table

Homeowner User Story:

As A	I Want To	So That	Acceptance Criteria
Homeowner	Register & Create Account	I can start using the system.	User provides name, email and password. System validates and assigns homeowner role.
	Login to the system	I can access my dashboard.	Valid email & password required
	Add smart devices	I can add home appliances	Can add devices like light, fan, AC
	Control devices remotely	I can operate devices from anywhere.	ON/OFF works instantly.
	Remove devices	I can manage unused devices	Can delete selected device.
	View device status	I can monitor my home.	Dashboard shows real-time status.
	Schedule automation	Devices run automatically	Can set time-based rules.
	Receive alerts	I know about emergencies	Notifications for fire, gas leak, intrusion.
	View security camera	I can monitor home remotely.	Live video available.
	Lock/unlock doors	I can ensure security	Smart lock responds instantly
	Add family members	Others can use system	Can invite users
	Add guests	Guests can access limited features	Temporary access provided

	Assign roles	Users get proper permissions	Can assign roles
	Verify user requests	I control who enters system	Can approve/reject requests
	Remove users	I can maintain security	Can delete users anytime
	Suspend users	I can stop suspicious activity	Can block user access
	Approve technician access	I can allow maintenance	Can grant/revoke access
	Monitor activity logs	I can track system usage	Logs show user actions
	Use voice commands	Hands-free control	Voice works correctly
	Receive system notifications	Stay updated	Notifications delivered
	Backup system data	Prevent data loss	Backup available
	Restore system data	Recover data	Restore works
	Logout from system	Secure session end	Session ends safely

Family Member User Story:

As A	I Want To	So That	Acceptance Criteria
Family Member	Login to the system	I can access home features	Valid credentials required
	View dashboard	I can see available devices	Dashboard shows permitted devices

	Control devices	I can use appliances easily	Can turn ON/OFF allowed devices
	View device status	I know which devices are active	Real-time status displayed
	Control room-based devices	Easy navigation	Devices grouped by rooms
	Use automation features	Daily tasks become easier	Can use predefined automation only
	Receive alerts	I stay informed about events	Gets notifications for important alerts
	Request additional access	I can use more features if needed	Can send access request to homeowner
	Follow access restrictions	System remains safe	Cannot access restricted settings
	View personal activity	I can track my actions	Can see own activity logs only
	Get notifications for device changes	Stay updated	Notified when device state changes
	Change Password	Ensure Security	Change the passwords if needed.

	Logout from system	Ensure account security	Session ends properly

Guest User Story:

As A	I Want To	So That	Acceptance Criteria
Guest	Access system temporarily	I can use basic devices	Temporary login/access provided
	Control selected devices	I feel comfortable during visit	Can control allowed devices only
	simple login	Easy access	login works properly
	Avoid restricted areas	System remains secure	Cannot access locks, cameras, settings
	Use room devices	Convenience	Can control lights/fans only
	Automatic logout	Security ensured	Session expires automatically
	No history access	Privacy maintained	Cannot view logs or past data

Technician User Story:

As A	I Want To	So That	Acceptance Criteria

Technician	Login to system	I can access tools	Valid technician credentials required
	Configure devices	Devices work properly	Can setup devices
	Run diagnostics	Identify problems	System shows error logs
	Access device logs	Understand issues	Logs are visible
	Perform maintenance	Ensure smooth operation	Devices function correctly after maintenance
	Logout	End session securely	Session ends properly

2.2 Requirements Traceability Matrix

2.2.1 Functional Requirement

	Requirement description
1	**Registration**
1.1	Enter username
1.2	Enter user Email
1.3	Enter Password
1.4	Confirm Password
1.5	Enter user Phone Number
1.6	Register button to send the register request
1.7	Clear button to clear the form
2	**Login**

2.1	Enter registered Email
2.2	Enter Password
2.3	Login button to login the system
2.4	Register button to go for registration.
2.5	Forgot password to recover password.
3	**Dashboard (user)**
3.1	Display User Info
3.2	Display Real-time Overview
3.3	Device Status Panel to control device
3.4	Update password
3.5	Logout Option
4	**Dashboard (Homeowner)**
4.1	Device Management to manage all device of owner room
4.2	Room Management to control room wise devices
4.3	User Management Panel to assign users.
4.4	User Access Control to restrict user actions.
4.5	Automation Panel to set automation
4.6	Security Panel for mode control and activity log.
4.7	Activity Logs of all users
5	**Add device (Homeowner)**

5.1	Device Input Capture
5.2	Device Type Selection
5.3	Assign Device to Room
5.4	Validate Device Information
5.5	Connect / Pair Device
5.6	Real-time Device Update
5.7	Error Handling (Connection Fail)
6	**Device ON/OFF Control**
6.1	ON/OFF Toggle Control
6.2	Real-time Command Execution
6.3	Device Status Update to the dashboard
6.4	Activity Logging
7	**Remove Device (Homeowner)**
7.1	Select the device
7.2	Delete Device Request
7.3	Delete Device from Database
7.4	Real-time Dashboard Update
7.5	Activity Logging
8	**Security Camera manage (Homeowner)**
8.1	View Live Camera Feed

8.2	Storage Management
8.3	Store recorded Video
8.4	Error Handling (Camera Offline)
9	**Guest Access (Homeowner)**
9.1	Limited Device Control
9.2	Auto Logout when Session Expires.
9.3	Store Activity Log
10	**Add family member (Homeowner)**
10.1	Send Invitation through Email/Phone.
10.2	Accept/Reject request
10.3	Assign Role
10.4	Give Role-Based Access Control
10.5	Store Activity Log
11	**Remove family member (Homeowner)**
11.1	Select Family Member
11.2	Delete member
11.3	Confirmation Prompt
11.4	Update Dashboard in Real-time
12	**Verify user request (Homeowner)**

12.1	Receive User Request
12.2	Check User Identity
12.3	Role-Based Permission Check
12.4	Approval / Rejection Decision
12.5	Notify User about Decision
13	**Technician Access Management (Homeowner)**
13.1	Receive Technician Access Request
13.2	Approve Technician Request
13.3	Reject Technician Request
13.4	Assign Maintenance Permission
13.5	Revoke Technician Access
14	**Automation Scheduling (Homeowner & Family Member)**
14.1	Create Automation Rule
14.2	Set Time Schedule
14.3	Edit Automation Rule
14.4	Delete Automation Rule
14.5	Enable/Disable Automation
14.6	Store Automation Activity Log
15	**Alert & Notification Management**
15.1	Receive Gas Leak Alert

15.2	Receive Fire Alert
15.3	Receive Intrusion Alert
15.4	Push Notification Delivery
15.5	View Notification History
15.6	Mark Notification as Read
16	**Energy Monitoring (Homeowner & Family Member)**
16.1	Display Daily Energy Usage
16.2	Display Monthly Energy Usage
16.3	Calculate Estimated Electricity Cost
16.4	Generate Energy Usage Report
17	**Smart Door Lock Management (Homeowner & Family Member)**
17.1	Lock Door
17.2	Unlock Door
17.3	Real-time Lock Status Update
17.4	Unauthorized Access Detection
17.5	Store Door Access Logs
18	**Forgot Password**
18.1	Enter Registered Email
18.2	Send OTP
18.3	Verify OTP

18.4	Reset Password
18.5	Confirm New Password
19	**Profile Management**
19.1	View Profile
19.2	Update User Information
19.3	Change password
19.4	Upload Profile Picture
19.5	Logout from the system.

2.2.2 Non-Functional Requirements

No	Requirement Description
1	**Performance**
1.2	The system shall load the homeowner's dashboard within 3 seconds under normal network conditions.
1.3	The system shall execute device control commands such as turning lights or fans ON/OFF within 1 second after the user taps the toggle.
1.4	The system shall update device status and automation schedules on the dashboard in real-time after any change is made.
1.5	The system shall deliver emergency alerts such as fire, gas leak, and intrusion notifications without noticeable delay.
2	**Usability**
2.1	The system shall provide a simple, clear, and user-friendly interface for homeowners, family members, guests, and technicians with consistent navigation across all screens.

2.2	The system shall use understandable labels, buttons, and messages so that non-technical users such as elderly family members can operate devices without training.
2.3	The system shall provide proper validation of messages when users enter incorrect or incomplete information during registration, login, or device setup.
2.4	The system shall maintain consistent navigation across registration, login, dashboard, device control, security, automation, and energy monitoring pages.
3	**Security**
3.1	The system shall protect all user accounts through secure login and authentication mechanisms for all roles including homeowners, family members, guests, and technicians.
3.2	The system shall store passwords using secure hashing techniques and shall never display passwords in plain text.
3.3	The system shall apply role-based access control so that guests cannot access security locks or cameras; family members cannot modify automation settings, and only homeowners have full administrative control.
3.4	The system shall encrypt all communication between the mobile application and the server to prevent unauthorized data interception.
4	**Reliability**
4.1	The system shall keep device state, user activity logs, automation schedules, and security data consistent after each successful operation.
4.2	The system shall handle failed device commands or lost network connections by showing appropriate error messages instead of crashing.
4.3	The system shall support regular data backups to prevent loss of important user profiles, device configurations, and activity logs.
4.4	The system shall maintain stable session management and automatically log out guest and technician sessions after a set period.
5	**Availability**
5.1	The system shall remain available to homeowners and family members with minimum downtime during regular operation.
5.2	The system shall allow users to access critical features such as device control, security monitoring, and alerts whenever they are logged in.

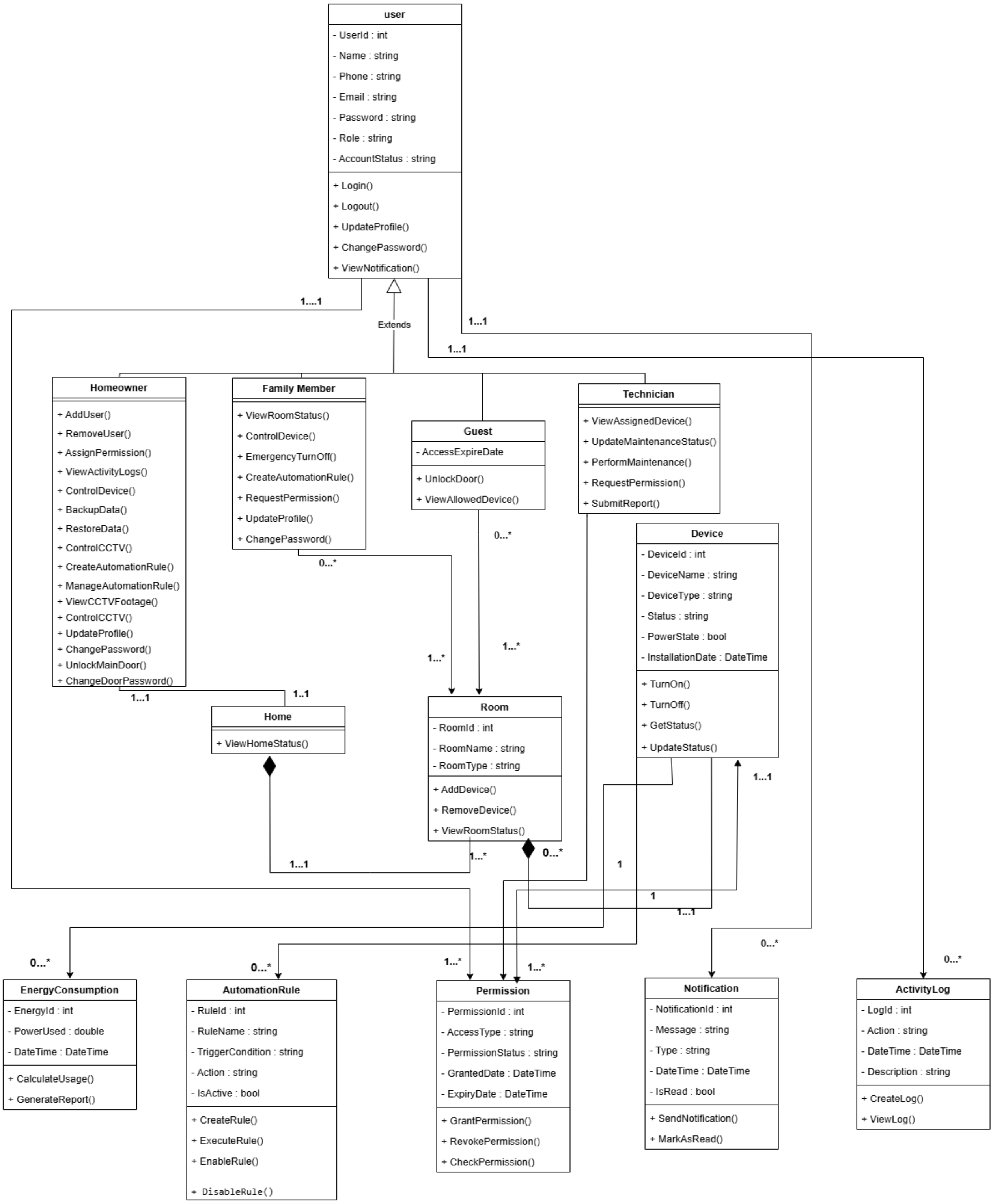
5.3	The system shall automatically reconnect to the network and synchronize device status after a brief internet or power outage.
5.4	The system shall notify users properly if a service or feature such as live camera feed is temporarily unavailable.
6	**Scalability**
6.1	The system shall support adding new smart devices, rooms, automation rules, and user accounts without changing the core system structure.
6.2	The system shall handle an increasing number of connected smart devices and household users as the home grows without degrading performance.
6.3	The database design shall support future expansion of device types, user roles, energy records, and activity logs.
6.4	The voice assistant module shall support future integration with advanced speech recognition and natural language processing technologies.
7	**Maintainability**
7.1	The system code shall be organized into separate modules such as user management, device control, security, automation scheduling, energy monitoring, and technician management.
7.2	The system shall allow developers to update features and fix issues in one module without affecting other unrelated modules.
7.3	The system shall follow clear naming conventions, inline documentation, and version control practices to support team collaboration.
7.4	The system shall be tested after each major update to ensure all existing features including device control, alerts, and role-based access continue to work properly.
8	**Compatibility**
8.1	The system shall be accessible as a mobile application on both Android and iOS platforms.
8.2	The system interface shall be fully responsive and adapt properly to different screen sizes, including smartphones and tablets.
8.3	The system shall display dashboards, device cards, alerts, and energy charts correctly across all supported devices.
8.4	The system shall maintain consistent behavior and appearance across supported mobile operating system versions.

3. SOFTWARE DESIGN

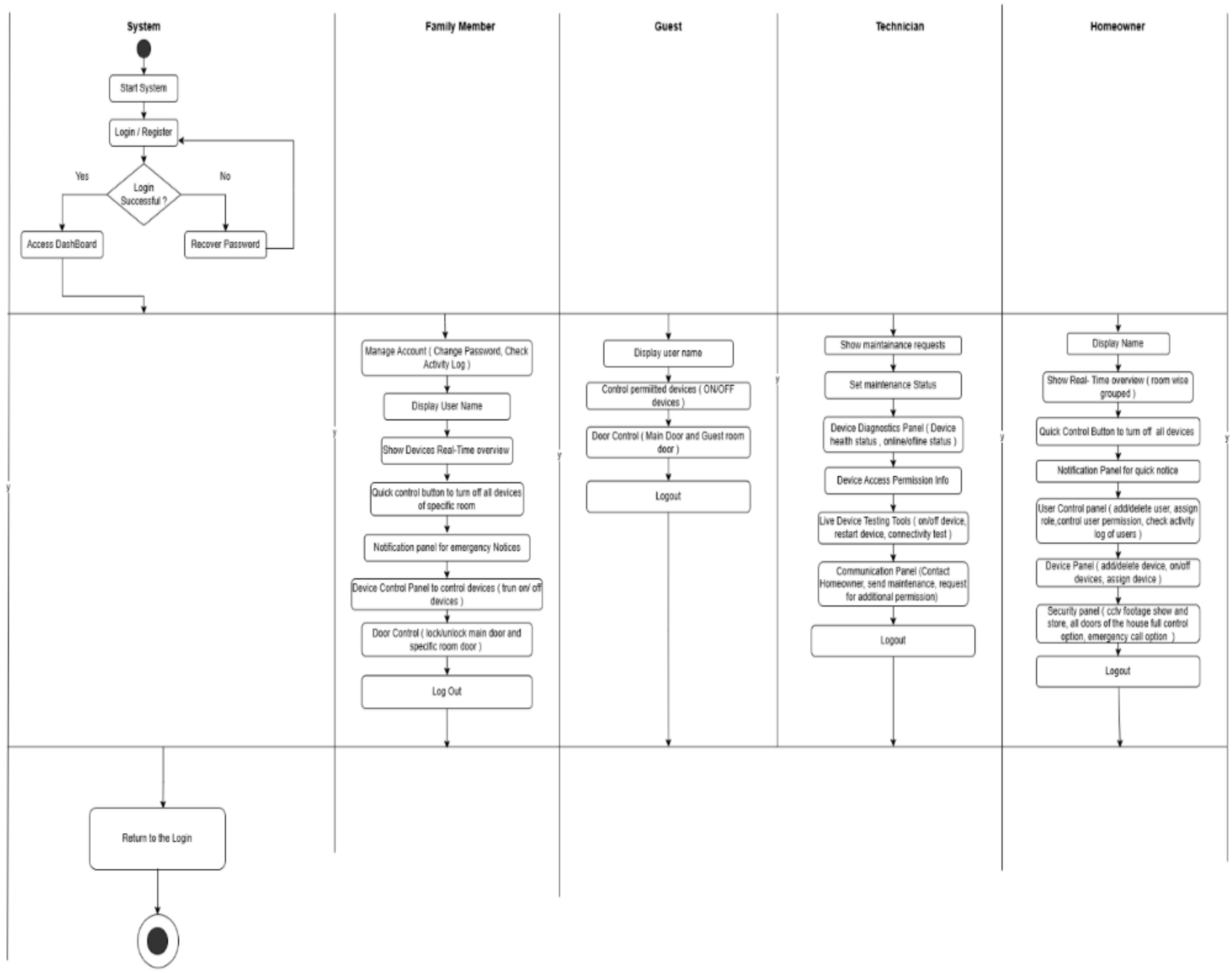
3.1 System Design

3.1.1 Use Case Diagram

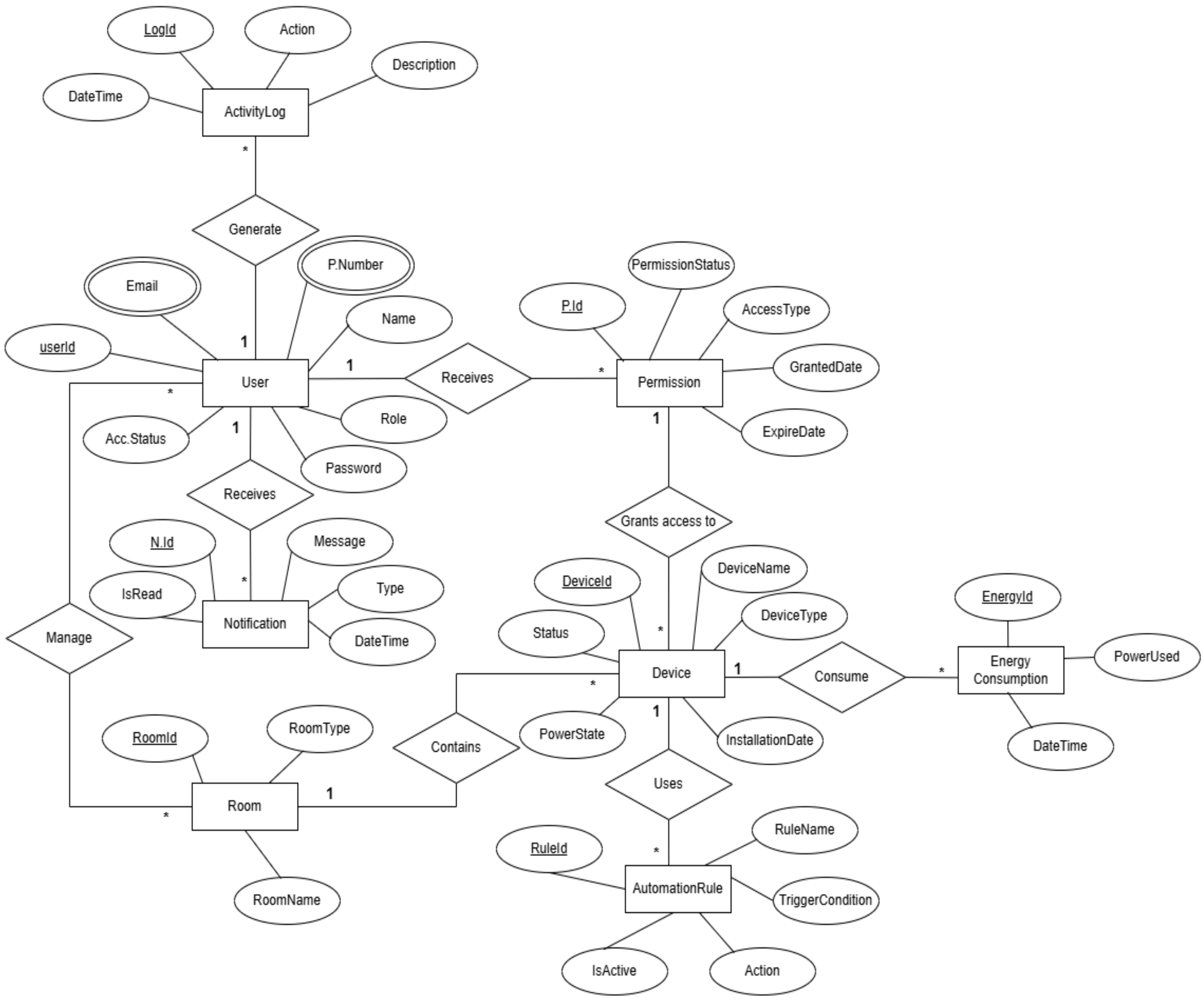
3.1.2 Class Diagram



3.1.3 Activity Diagram



3.1.4 ER Diagram



6.CONCLUSION:

The Smart Home Management System (Smart Home+) is designed to provide a smarter, safer, and more efficient way of managing household activities through a centralized mobile application. Traditional homes often face common challenges such as energy waste, limited remote access, weak security monitoring, and difficulty in managing multiple devices efficiently. This system addresses these issues by integrating essential smart home functionalities into a single platform. The proposed system enables homeowners to remotely control appliances such as lights, fans, and air conditioners, monitor security devices, manage smart locks, receive emergency alerts, and track electricity consumption. In addition, the system supports multiple user roles including homeowners, family members, guests, and technicians, ensuring controlled access and better management of responsibilities. Advanced features such as voice assistant support and security monitoring further improve user convenience and home safety. The selection of the Incremental Process Model makes the development process more organized and practical by dividing the system into smaller functional modules. This approach simplifies development, testing, debugging, and future expansion while reducing overall project complexity. Overall, Smart Home+ demonstrates how modern mobile technology can improve home automation, strengthen security, reduce energy consumption, and enhance the daily living experience for users in an intelligent and convenient way.